

CSP650 FINAL YEAR PROJECT Structuring Report

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(Source :
Adapted from (1) Liv Jonassen & Elizabeth Tomchak, The Robert Gordon Univ. Aberdeen,
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Section in the report

- Draft your table of contents based on guideline given from faculty
 - Essentials of Computing Sciences
- Identify chapter for the report
 - Identify sub-sections for every chapter
 - Identify sub-sub-sections
- Search example dissertation from other university from online database to get an idea
- Discuss the draft of TOC with your supervisor

Table of contents - Sample A

SUPERVISOR'S APPROVAL

STUDENT'S DECLARATION

ACKNOWLEDGEMENT

ABSTRACT

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

CHAPTER ONE: INTRODUCTION

CHAPTER TWO: LITERATURE REVIEW

CHAPTER THREE: RESEARCH METHODOLOGY

CHAPTER FOUR: RESULTS AND ANALYSIS

CHAPTER FIVE: CONCLUSION AND FUTURE WORK

References

Appendices

Table of contents - Sample B

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TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

CHAPTER ONE: INTRODUCTION

CHAPTER TWO: LITERATURE REVIEW

CHAPTER THREE: RESEARCH METHODOLOGY

CHAPTER FOUR: RESEARCH DESIGN AND IMPLEMENTATION

CHAPTER FIVE: RESULTS AND ANALYSIS

CHAPTER SIX: CONCLUSION AND FUTURE WORK

References

Appendices

Chapter 1 - Introduction

- Objective
 - to present the motivation of the project
 - to provide the context, key terms, and concept so the reader can understand the domain
 - to present the research question, objective, scope and significant of the study
 - to describe the overall structure of the project

Chapter 1 – Content

1.1 Introduction

1.2 Background of Study– what is the content in the background?

1.3 Problem statement – must have ref. and citations

1.4 Research questions

1.5 Objectives –

- To identify.../To investigate... /To analyse...
- To design.../To develop.../To construct...

1.6 Project Scope

1.7 Project Significance

1.8 Report organization

Chapter 2 - Literature Review

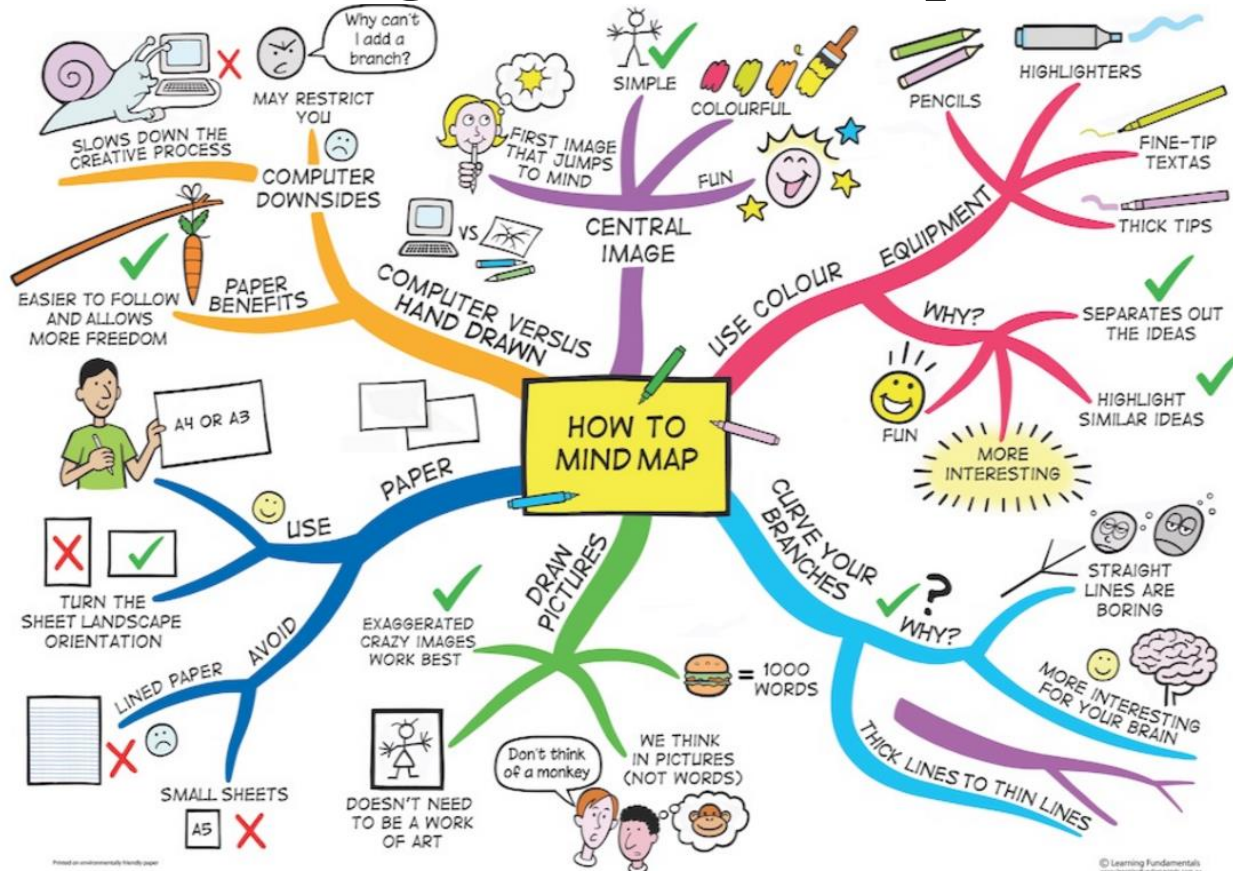
- Objective
 - To justify the needs of the project – your project is worth
 - To discuss and critically evaluate past and current studies in your area
 - To provide reader and other researchers with a starting point from which they can understand your project

Chapter 2 - Tips Content

- The content varies according to project area, 3-4 mains sections should be sufficient
- Organise the content from general to specific about your project area
 - E.g: 2.1. Introduction, 2.2.Intelligent System , 2.2.1 Expert System, 2.2.2.Fuzzy Expert System, 2.2.3 Case-based Reasoning.
- Highlight your project area (specific technique/model) that gives in depth details about your project
- Highlight the literatures that support your problem statement in and objective stated in Chapter 1
- Concrete your statements/arguments with citations and make use of reference tools (Mendeley & Endnotes)
- Find the latest references (last 3-5 years), mix of journals/proceedings/books/technical report etc.

Chapter 2 - Tips Content (cont.)

- Mind map of your project area to check whether you are covering most of the important things



Chapter 2 - Suggested Content

- 2.1 Introduction
- 2.2 Overview of the selected area
 - 2.2.1 Definition of the concept
 - 2.2.2 Issues and Challenges
- 2.3 Techniques in the area
 - 2.3.1 Technique A
 - 2.3.2 Technique B
 - 2.3.3 Technique C
 - 2.3.4 Conclusion + Comparison of the techniques
- 2.4 Selected Technique
 - 2.4.1 Past studies on the selected area
 - 2.4.2 Comparison of the results that apply that technique
 - 2.4.3 Conclusion + highlight the gap/limitation of the technique
- 2.5 Summary

Chapter 2 - How to write

- Be creative and write in different style
- Be selective - remember it is not a report that you have to write/summarize all the papers and books that you have read
- Do not dedicate a page or paragraph to each article or merely reporting on their content
- See examples

Example 1 - citation

- **The definition by** John *et al.* (1994) emphasises feature selection (FS) to improve the prediction accuracy or to decrease the size of the structure by selecting a subset of features without significantly decreasing prediction accuracy of the classifier built using only the selected features.
- **In contrast,** Liu & Yu (2005) defined FS as a process that finds the minimal size features subset from the raw features whereby the ideal size of the feature subset is based on some evaluation criteria. **Similarly,** Guyon (2008) defined FS as making good predictions with as few features as possible.

Example 2 - citation

- Feature Selection (FS) and Feature Transformation (FT) are among the pre-processing methods that are commonly used in the literature. These two methods basically aim to change the data into a simpler dimension that enables the machine learning algorithms to learn faster. Usually, ML algorithms would degrade in performance when faced with many features that are not necessary for rule discovery (Jain et al. 2000; Zhong & Dong 2001; Liu et al. 2002; Liu & Yu 2005; Saeys et al. 2007; Luka & Zoran 2010).

Example 3 - citation

- FS methods have been applied in various fields such as in **bioinformatics** (Al-Shahib et al. 2005; Hauskrecht et al. 2006; Chuang et al. 2008; Nemati et al. 2009; Unler & Murat 2010), **signal processing** (Guldogan & Gabbouj 2008; Nemati et al. 2008), **text categorisation** (Shang et al. 2007; Aghdam et al. 2009), **image retrieval** (Song et al. 2010), and **intruder detection** (Mukkamala & Sung 2005).
- KD is defined by Fayyad et al. as a “*nontrivial process of identifying valid, novel, potentially useful and ultimately understandable patterns in data*” (1996: 40) and cited in Vanschoren (2010).

Chapter 3 : Research Methodology

- Objective

- To describe the overall process flow of the research (framework of the research, research structure)
- To describe the techniques and procedure (steps) used in research; eg. Steps in developing the algorithm, steps in pre-processing the dataset
- To explain the data, materials, subjects, tools and equipment used in research
- To explain how the experiments are designed
- Generally to describe:

How the project/research is carried out?

Chapter 3 – Suggested Content

- 3.1 Introduction
- 3.2 Theoretical Study
 - 3.2.1 Preliminary Study (produce project structure/framework)
 - 3.2.2 Knowledge Acquisition (interview results, gap in literature)
- 3.3 Empirical Study
 - 3.3.1 Data Collection
 - 3.3.2 Data Pre-processing
 - 3.3.3 Data Analysis
- 3.4 System Design
 - 3.4.1 Database Design
 - 3.4.2 Interface Design
 - 3.4.3 Architecture Design
- 3.5 System Development
 - 3.5.1 System Flow
 - 3.5.2 System Testing and Evaluation
 - 3.5.3 Hardware and Software requirements
- 3.6 Summary

Chapter 3 - Tips

- Provide enough detail for replication of your work.
- Order procedures chronologically.
- Use past tense to describe what you did.
- Don't mix results with procedure.

Chapter 4: Results and Analysis

- To be discussed in next topic

Chapter 5: Conclusion & Future Works

- To be discussed in next topic

References

- List all references that you have cited in the project
- Make sure you cite all the references that you have listed in this section

Appendices

- List all related documents to support your project
- Example:
 - Questionnaire
 - Survey results
 - Sample of data – before pre-processing
 - Sample of data – after pre-processing
 - Details results of findings